

Method of variation of parameters.

• First order diff Eqn :- $y' + p(x)y = r(x)$

$p(x), r(x)$ are continuous functions on interval I
Then we find solution by using method of variation of parameters.

Step I :- $y' + p(x)y = 0$ (corresponding Homogeneous Eqn)

→ find solution of this Homogeneous Eqn.

Let $y_1(x)$ is solution of homogeneous Eqn.

• By method of variation of parameters, we get

$$y_p(x) = y_1(x) \int \frac{r(x)}{y_1(x)} dx \text{ is solution of Non homo Eqn}$$

Complete solution is

$$y = C_1 y_1(x) + y_p(x)$$

• Let $y_p(x) = u(x) y_1(x)$

$$y_p'(x) = u(x) y_1'(x) + u'(x) y_1(x)$$

$$y' + p(x)y = r(x)$$

Using given Eqn, we get

$$u(x) y_1'(x) + u'(x) y_1(x) + p(x) u(x) y_1(x) = r(x)$$

$$u(x) \left[\underbrace{y_1'(x) + p(x) y_1(x)}_{=0} \right] + u'(x) y_1(x) = r(x)$$

$$\Rightarrow u'(x) = \frac{r(x)}{y_1(x)}$$

$$\Rightarrow u(x) = \int \frac{r(x)}{y_1(x)} dx$$

$$y_p(x) = u(x) y_1(x) = y_1(x) \int \frac{r(x)}{y_1(x)} dx$$

Example

Solve

$$(x+4) \frac{dy}{dx} + 3y = 3 \Rightarrow \frac{dy}{dx} + \frac{3}{(x+4)} y = \frac{3}{(x+4)}$$

Homogeneous Eqn :- $\frac{dy}{dx} + \frac{3}{(x+4)} y = 0$

$$\Rightarrow \frac{dy}{dx} = \frac{-3y}{(x+4)} \Rightarrow \int \frac{dy}{y} = \int \frac{-3 \cdot dx}{x+4}$$

$$\Rightarrow \log y = -3 \log(x+4) + \log C$$

$$\Rightarrow y = \frac{C}{(x+4)^3} \quad y_1(x) = \frac{1}{(x+4)^3}$$

By method of variation of parameters.

$$y_p(x) = y_1(x) \int \frac{r(x)}{y_1(x)} dx$$

$$y_p(x) = \frac{1}{(x+4)^3} \int \frac{3}{(x+4)} \cdot (x+4)^3 dx$$

$$y_p(x) = \frac{1}{(x+4)^3} \cdot \frac{3 (x+4)^3}{3} = 1$$

Complete solution :- $y = \frac{C}{(x+4)^3} + 1$

• Second order linear Equation

$$y'' + p(x) y' + q(x) y = r(x)$$

$p(x), q(x), r(x) \rightarrow$ continuous on some interval.

$$y(x) = -y_1 \int y_2 \frac{r(x)}{W} dx + y_2 \int \frac{y_1 r(x)}{W} dx$$

$$y_p(x) = -y_1 \int \frac{y_2 r(x)}{w} dx + y_2 \int \frac{y_1 r(x)}{w} dx$$

where y_1 and y_2 are basis of solution of corresponding homogeneous Eqn, w is Wronskian determinant of y_1 and y_2 .

$$w(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Example:- Solve $y'' + n^2 y = \sec(nx)$

Solution:- Homo Eqn $y'' + n^2 y = 0$

$$(D^2 + n^2)y = 0$$

Ch Eqn

$$r^2 + n^2 = 0 \Rightarrow r^2 = -n^2$$

$$\Rightarrow r = \pm ni$$

$$y_{\text{hom}} = C_1 \cos(nx) + C_2 \sin(nx)$$

$$y_1 = \cos(nx)$$

$$y_2 = \sin(nx)$$

• Wronskian of y_1 and y_2

$$w(y_1, y_2) = \begin{vmatrix} \cos(nx) & \sin(nx) \\ -n \sin(nx) & n \cos(nx) \end{vmatrix} = n \cos^2(nx) + n \sin^2(nx) = n$$

$$y_p(x) = -y_1 \int \frac{y_2 r(x)}{w} dx + y_2 \int \frac{y_1 r(x)}{w} dx$$

$$I_1 = \int \frac{y_2 r(x)}{w} dx = \int \frac{\sin(nx) \cdot \sec(nx)}{n} dx$$

$$= \frac{1}{n} \int \frac{\sin(nx)}{\cos(nx)} dx$$

$$\begin{aligned} \cos(nx) &= z \\ -n \sin(nx) dx &= dz \\ \sin(nx) dx &= \frac{-dz}{n} \end{aligned}$$

$$\begin{aligned} &= \frac{-1}{n^2} \int \frac{dz}{z} = \frac{-1}{n^2} \log z \\ &= \frac{-1}{n^2} \log(\cos(nx)) \end{aligned}$$

$$I_2 = \int \frac{y_2(x)}{w} dx = \int \frac{\cos(nx) \sec(nx)}{n} dx$$

$$= \frac{1}{n} \int dx = \frac{x}{n}$$

$$y_p(x) = -y_1 I_1 + y_2 I_2 = -\cos(nx) \cdot \frac{-1}{n^2} \log(\cos(nx)) + \sin(nx) \cdot \frac{x}{n}$$

$$y_p(x) = \frac{\cos(nx) \log(\cos(nx))}{n^2} + \frac{x}{n} \sin(nx)$$

Complete solution

$$y(x) = c_1 \cos(nx) + c_2 \sin(nx) + y_p(x).$$

• Consider the linear diff Eqn

$$y^n + p_{n-1} y^{n-1} + p_{n-2} y^{n-2} + \dots + p_1 y + p_0 = r(x)$$

$p_{n-1}, p_{n-2}, \dots, p_1, r(x) \rightarrow$ continuous function on some interval I .

$$y_p(x) = y_1(x) \int \frac{w_1(x)}{w(x)} r(x) dx + y_2(x) \int \frac{w_2(x)}{w(x)} r(x) dx + \dots$$

$$- + y_n(x) \int \frac{w_n(x)}{w(x)} r(x) dx$$

where $y_1, y_2, \dots, y_n$ are " n " L.I. solution of corresponding homogeneous Eqn. $w(x) = w(y_1, y_2, \dots, y_n)$

$w_j(x) \rightarrow$ obtained from $w(x)$ by replacing j th column with $(0, 0, \dots, 0, 1)^T$.

$$\underline{n=2} \quad y_p(x) = -y_1(x) \int \frac{y_2(x)}{w} r(x) dx + y_2(x) \int \frac{y_1(x)}{w} r(x) dx$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$w_1 = \begin{vmatrix} 0 & y_2 \\ 1 & y_2' \end{vmatrix} = -y_2, \quad w_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & 1 \end{vmatrix} = y_1(x)$$

Example :- Solve $y^{(3)} + y^{(1)} = \operatorname{cosec} x$

Solution :- Homogeneous Eqn
Ch Eqn

$$y^{(3)} + y^{(1)} = 0$$

$$x^3 + x = 0 \Rightarrow x(x^2 + 1) = 0$$

$$\Rightarrow x = 0, +i, -i$$

$$y_{\text{hom}} = C_1 e^{0x} + C_2 \cos x + C_3 \sin x$$

$$y_1(x) = 1, \quad y_2(x) = \cos x, \quad y_3(x) = \sin x$$

$$W(y_1, y_2, y_3) = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{vmatrix} = \begin{vmatrix} -\sin x & \cos x \\ -\cos x & -\sin x \end{vmatrix}$$

$$= +\sin^2 x + \cos^2 x = 1$$

$$w_1 = \begin{vmatrix} 0 & \cos x & \sin x \\ \dots & \dots & \dots \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x$$

$$W_1 = \begin{vmatrix} 0 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 1 & -\cos x & -\sin x \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$W_2 = \begin{vmatrix} 1 & 0 & \sin x \\ 0 & 0 & \cos x \\ 0 & 1 & -\sin x \end{vmatrix} \quad W_3 = \begin{vmatrix} 1 & \cos x & 0 \\ 0 & -\sin x & 0 \\ 0 & -\cos x & 1 \end{vmatrix}$$

$$* W_2 = \begin{vmatrix} 0 & \cos x \\ 1 & -\sin x \end{vmatrix} = -\cos x$$

$$W_3 = \begin{vmatrix} -\sin x & 0 \\ -\cos x & 1 \end{vmatrix}$$

$$W_3 = -\sin x$$

$$I_1 = \int \frac{W_1(x)}{W(x)} \cdot x(x) dx = \int \operatorname{cosec} x dx = \log | \operatorname{cosec} x - \cot x |$$

$$I_2 = \int \frac{W_2}{W} x(x) dx = \int \frac{-\cos x}{1} \cdot \operatorname{cosec} x dx = - \int \frac{\cos x}{\sin x} dx$$

$$\sin x = z \quad \Rightarrow \quad \cos x dx = dz$$

$$= - \int \frac{dz}{z} = -\log z = -\log(\sin x)$$

$$I_3 = \int \frac{W_3}{W} x(x) dx = \int \frac{-\sin x}{1} \cdot \operatorname{cosec} x dx = \int -1 \cdot dx = -x$$

$$y_p(x) = y_1(x) I_1 + y_2(x) I_2 + y_3(x) I_3$$

$$y_p(x) = 1 \cdot \log | \operatorname{cosec} x - \cot x | + \cos x \cdot (-\log(\sin x)) + \sin(x) \cdot (-x)$$

Complete solution is

$$y = C_1 + C_2 \cos x + C_3 \sin x + \log |\operatorname{cosec} x - \cot x| - \cos x \log(\sin x) - x \sin x.$$

o solve $x^2 y'' - xy' - 3y = x^2 \log x$

$$\begin{aligned} x &= e^t \\ xy' &= \frac{dy}{dt} \\ x^2 y'' &= \frac{d^2 y}{dt^2} - \frac{dy}{dt} \end{aligned}$$

$$D = \frac{d}{dt}$$

* $((D^2 - D) - D - 3)y = e^{2t} \cdot t$
 $(D^2 - 2D - 3)y = e^{2t} \cdot t$

Ch eqn $x^2 - 2x - 3 = 0$ $x = 3, -1$

$$y_{\text{hom}} = C_1 e^{3t} + C_2 e^{-t}$$

P.I. $\frac{1}{D^2 - 2D - 3} \cdot e^{2t} \cdot t = e^{2t} \frac{1}{(D+2)^2 - 2(D+2) - 3} \cdot t$

$$= e^{2t} \frac{1}{D^2 + 4D + 4 - 2D - 4 - 3} \cdot t = e^{2t} \frac{1}{D^2 + 2D - 3} \cdot t$$

$$= \frac{e^{2t}}{-3} \cdot \left(1 - \left(\frac{D^2 + 2D}{3}\right)\right)^{-1} t$$

$$= \frac{e^{2t}}{-3} \left(1 + \frac{D^2}{3} + \frac{2}{3}D\right) t = \frac{e^{2t}}{-3} \left(t + \frac{2}{3}\right)$$

$$= \frac{e^{2t}}{-3} \left(1 + \frac{D^2}{3} + \frac{2}{3}D \right) t = \frac{e^{2t}}{-3} \left(t + \frac{2}{3} \right)$$

Complete solution

$$y = C_1 e^{3t} + C_2 e^{-t} - \frac{1}{3} \cdot t e^{2t} - \frac{2}{9} e^{2t}$$

$$e^t = x \Rightarrow t = \log x$$

$$y = C_1 x^3 + \frac{C_2}{x} - \frac{1}{3} \log x \cdot x^2 - \frac{2}{9} \cdot x^2$$